

GEOFFERY EYRAM AGORKU

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RESEARCH PROFILE

PhD student in Civil Engineering (Transportation Engineering) and Doctoral Academy Fellow specializing in inland waterway and multimodal freight systems. My research integrates machine learning, computer vision, Automatic Identification System (AIS)-based vessel analytics, and geospatial modeling to develop operational tools for infrastructure planning, freight network analysis, and real-time transportation monitoring. I have collaborated with the U.S. Army Engineer Research and Development Center (ERDC), U.S. Army Corps of Engineers (USACE), on applied analytics for national freight systems, with work spanning multimodal routing, commodity flow visualization, vessel and barge detection, and infrastructure resilience. My work has been recognized through peer-reviewed publications, technical reports, national and regional awards, and presentations at the Transportation Research Board (TRB) Annual Meeting, the Missouri Valley District Institute of Transportation Engineers (MOVITE) Spring Meeting, Smart Rivers, and other invited forums.

Future work focuses on developing scalable, AI-driven maritime and multimodal transportation monitoring systems for U.S. inland waterways, advancing data-driven decision-making for national infrastructure planning, real-time situational awareness, and transportation resilience.

RESEARCH IMPACT & CONTRIBUTIONS

- Developed multimodal data-fusion frameworks integrating AIS trajectory data, traffic camera imagery, and satellite observations for inland waterway monitoring and vessel characterization.
- Designed and implemented machine learning models for barge detection and tow size estimation, achieving up to 96% detection accuracy and strong predictive performance across real-world datasets.
- Contributed to U.S. Army Corps of Engineers (USACE) and U.S. Army ERDC research efforts on national freight systems, with outputs published as federally archived technical reports and applied to infrastructure planning and operational analytics.
- Produced decision-support tools and analytics for evaluating freight disruptions, supporting infrastructure planning and operational efficiency in inland waterway systems.
- Advanced research at the intersection of transportation systems, computer vision, and geospatial analytics, with applications to national infrastructure monitoring and policy-relevant decision-making.

RESEARCH INTERESTS

Inland waterway transportation systems; multimodal freight analytics; machine learning for transportation; computer vision for vessel and barge detection; AIS-based vessel behavior modeling; geospatial analysis and GIS; transportation resilience and infrastructure disruption analysis; freight decision-support systems.

EDUCATION

University of Arkansas, Fayetteville, AR **Aug. 2024 – May 2028 (expected)**

Doctor of Philosophy, Civil Engineering (Transportation Engineering), GPA: 3.8/4.0

- *Advisor:* Sarah Hernandez, Ph.D., P.E.
- *Research focus:* Inland waterway and multimodal freight analytics using AIS, machine learning, computer vision, and geospatial modeling.

University of Arkansas, Fayetteville, AR **Jan. 2023 – Aug. 2024**

Master of Science, Civil Engineering (Transportation Engineering), GPA: 3.9/4.0

Kwame Nkrumah University of Science and Technology (KNUST), Kumasi, Ghana Aug. 2017 – Nov. 2021

Bachelor of Science, Civil Engineering | First Class Honors (CWA: 76.6/100); Excellent Student Award 2021

PROFESSIONAL EXPERIENCE

Civil Engineer in Training, Ghana Highway Authority, Ho, Ghana Oct. 2021 – Nov. 2022

- Used Civil 3D to develop road alignments, cross-sections, and profiles for more than 30 miles of roadway, accounting for topography and drainage considerations.
- Built custom software to support compaction and grading analysis and improve quality-control efficiency during construction.

Civil Engineering Intern, Ghana Highway Authority, Ho Municipal, Volta, Ghana Aug. 2020 – Dec. 2020

- Conducted materials and geotechnical testing for pavement and roadway evaluation, including field density tests, California Bearing Ratio (CBR), particle size distribution, Atterberg limits, compaction tests, and swell tests, to support quality control and pavement assessment.
- Performed site inspections, road tests, technical reporting, and research activities; developed an application to automate grading and Atterberg limits calculations/reporting, reducing manual errors and improving reporting efficiency, while helping train new interns.

Civil Engineering Intern, Munck Civil Engineering, Doninga, Ghana May 2019 – Jul. 2019

- Conducted materials testing for soil, concrete, asphalt, and aggregates as part of the Danida Bridges project, which involved seven bridges and associated roadway works in Northern Ghana.
- Performed routine site inspections to monitor construction progress, material quality, and safety compliance.

Civil Engineering Intern, Ghana Highway Authority, Cape Coast Municipal, Central, Ghana Jun. 2017 – Aug. 2017

- Supported the Materials Division through field and laboratory testing for roadway and pavement assessment, including field densities, CBR, particle size distribution, Atterberg limits, compaction testing, and related road tests.
- Assisted with site inspections, report writing, research activities, and intern training, contributing to materials evaluation workflows and improving the reliability of field and laboratory reporting.

PUBLICATIONS

Peer-Reviewed Journal Articles (Published)

Geoffery Eyram Agorku, Sarah Hernandez, Maria Falquez, Subhadipto Poddar, Kwadwo Amankwah-Nkyi.

“Real-Time Barge Detection Using Traffic Cameras and Deep Learning on Inland Waterways.” Transportation Research Record: Journal of the Transportation Research Board, 2679(2), 703–720, 2025. DOI:

10.1177/03611981241263574

Manuscripts Under Review

Geoffery Eyram Agorku, Sarah Hernandez, Hayley Hames, Cade Wagner. “Predicting Barge Tow Size on Inland Waterways Using Vessel Trajectory Derived Features: Proof of Concept.” arXiv:2510.23994 [cs.LG], 2025.

DOI: 10.48550/arXiv.2510.23994

Geoffery Eyram Agorku, Sarah Hernandez, Hayley Hames, Cade Wagner. “Enhancing Maritime Domain Awareness on Inland Waterways: A YOLO-Based Fusion of Satellite and AIS for Vessel Characterization.”

arXiv:2510.11449 [cs.CV], 2025. DOI: 10.48550/arXiv.2510.11449

Preprints & Working Papers

Geoffery Eyram Agorku, Sarah Hernandez, Maria Falquez, Subhadipto Poddar, Shihao Pang. “Predicting Barge Presence and Quantity on Inland Waterways Using Vessel Tracking Data: A Machine Learning Approach.”

Preprint available on arXiv: arXiv:2501.00615. DOI: 10.48550/arXiv.2501.00615

Shihao Pang, Sarah Hernandez, Suman Mitra, **Geoffery Eyram Agorku**. “Identifying Waterway Traffic Flow Patterns Using Modified Clustering.”

Geoffery Eyram Agorku, Divine Agbobli, Vuban Chowdhury, Kwadwo Amankwah-Nkyi, Adedolapo Ogungbire, Portia Ankamah Lartey, Armstrong Aboah. “Real-Time Helmet Violation Detection Using YOLOv5 and Ensemble Learning.” arXiv:2304.09246 [cs.CV], 2023. DOI: 10.48550/arXiv.2304.09246
[Demonstrates broader expertise in computer vision applications for transportation safety.]

Technical Reports

Magdalena Asborn, Kenneth Ned Mitchell, Dustin Brown, Steven Peterson, **Geoffery Eyrar Agorku**, Mark Cowan. “*Major Freight Corridors in the US: Mapping of Commodity Flows on Waterborne, Rail, and Truck Networks.*” Engineer Research and Development Center, Navigation Systems Research Program, Final Technical Report, January 2026.

Hernandez, S., Mitra, S., **Agorku, Geoffery Eyrar**, Ashqui-Lopez, L., and Zolali, M.

“*Evaluation of Impacts Due to a Bridge Closure: A Case Study of the Mississippi River Bridges in Arkansas.*”

Final Report TRC2303, University of Arkansas, 2025. Published in the U.S. Department of Transportation

Repository and Open Science Access Portal (ROSA P). Available at: <https://rosap.ntl.bts.gov/view/dot/90534>

INVENTION DISCLOSURE

- 2026-028: Integrated Satellite and AIS-Based System for Barge and Vessel Detection with Predictive Tow Size Modeling. Invention disclosure filed; provisional patent in progress.

PROFESSIONAL SERVICE & SCHOLARLY CONTRIBUTIONS

- Reviewer for peer-reviewed journals including Transportation Research Record (TRB flagship journal), Ocean Engineering, Frontiers in Artificial Intelligence, and Transportation Research Interdisciplinary Perspectives, spanning transportation systems, maritime engineering, and AI applications.
- Completed 15+ peer reviews across high-impact journals and conferences, contributing to the advancement of research in transportation systems and AI-driven infrastructure monitoring relevant to national-scale applications.
- Invited as a peer reviewer based on demonstrated subject-matter expertise in machine learning applications for transportation systems, reflecting recognition within the academic research community.

TECHNICAL SYSTEMS & DATA PLATFORMS

- Multimodal freight routing and network analysis system developed in collaboration with USACE/ERDC, integrating AIS-derived waterway performance with rail and highway cost structures to support national freight infrastructure planning and disruption analysis.
- AIS-based vessel behavior modeling pipeline for trajectory processing, stop detection, and trip prediction, supporting large-scale analysis of inland waterway operations and traffic flow dynamics.
- Real-time barge detection system using computer vision and deep learning, achieving high-accuracy vessel identification from traffic camera imagery and enabling automated monitoring of inland waterway activity.
- Satellite–AIS data fusion framework for vessel detection and characterization, integrating remote sensing and trajectory data to support scalable monitoring of maritime activity across inland waterways.
- Infrastructure resilience and freight disruption analysis platform combining AIS data, commodity flows, and simulation-based modeling to evaluate economic impacts of bridge closures and support data-driven infrastructure planning.

RESEARCH EXPERIENCE

Graduate Civil Engineering Research Intern (Federal), U.S. Army Engineer Research and Development Center (ERDC), U.S. Army Corps of Engineers (USACE), Vicksburg, MS Aug. 2024 – Dec. 2024

Multimodal Freight Routing and Network Analytics

- Developed a multimodal freight routing framework integrating AIS-derived waterway performance with highway and rail cost structures to support system-level analysis of U.S. freight networks, enabling many-to-many routing and comparative scenario evaluation across transportation modes.
- Enhanced optimization and interface capabilities by introducing disruption-based scenario analysis and performance metrics, enabling assessment of network resilience and rerouting strategies under infrastructure failures or link disruptions relevant to national freight operations.

Freight Corridor Identification and Commodity Flow Visualization

- Developed data integration algorithms combining waybill and multimodal datasets to assign commodity

flows to specific rail and waterway network links, enabling network-level characterization of U.S. freight movement patterns.

- Designed geospatial visualization tools to map commodity flows across waterborne, rail, and truck systems, supporting identification of critical freight corridors and improving understanding of inland waterways' role in national logistics and supply chains.

Computer Vision for River Information Systems

- Defined data collection strategies and constructed image datasets for modeling barge configurations, supporting development of real-time river information systems for inland waterway monitoring.
- Developed image-based computer vision models to identify barge configurations, enabling automated situational awareness and real-time information dissemination for waterway operations and management.

Graduate Research Assistant, University of Arkansas, Fayetteville, AR

Jan. 2023 – Present

Machine Learning and Computer Vision for Inland Waterway Systems

- Developed a novel multimodal data integration framework linking AIS vessel trajectories with traffic camera imagery to generate labeled datasets, enabling scalable machine learning applications for vessel monitoring and classification in inland waterway systems.
- Built predictive models using AIS-derived operational features (e.g., vessel speed and maneuverability), achieving F1 scores up to 0.932 for barge detection and 0.886 for tow size estimation, supporting automated monitoring of freight movement and operational analysis.
- Designed and implemented a deep learning model for real-time barge detection from traffic camera feeds, achieving 96% accuracy, with results published in Transportation Research Record, contributing to advancements in AI-enabled infrastructure monitoring.
- Developed a satellite-AIS computer vision pipeline for detecting vessels and infrastructure features (bridges, docks, mooring areas), enabling large-scale, remote monitoring of inland waterway assets and vessel activity.

Transportation Data Science and Vessel Behavior Modeling

- Developed a vessel stop-detection algorithm using AIS data, achieving 90% accuracy, enabling identification of congestion points and operational inefficiencies in inland waterways.
- Implemented predictive modeling approaches for waterway trip analysis using vessel trajectory data, contributing to improved understanding of transportation behavior and network performance.
- Applied clustering and data-driven analytics to characterize traffic flow patterns and commodity movement, supporting transportation planning, system optimization, and policy-informed decision-making.

Infrastructure Resilience and Freight Decision Support

- Developed analytical methods to quantify the economic impacts of Mississippi River bridge closures using AIS-derived delay metrics, contributing to a U.S. Department of Transportation technical report and enabling data-driven evaluation of freight disruption risks in a nationally critical inland waterway corridor.
- Contributed to a federally archived technical report published on the U.S. Department of Transportation's ROSA P platform, translating AIS-based analytics into actionable insights for infrastructure resilience and national freight system risk assessment.
- Designed and implemented interactive decision-support dashboards for visualizing multimodal disruption scenarios, supporting scenario-based infrastructure planning and resilience analysis for transportation agencies.
- Developed simulation and commodity-flow-driven analytical tools to assess system-level vulnerability and mitigation strategies, enabling data-informed infrastructure investment prioritization and risk management.

Additional Applied Research Projects

- Designed and beta-tested questionnaires to evaluate rider satisfaction and operational efficiency for Razorback Transit in Fayetteville, Arkansas.
- Used ArcGIS spatial analysis and modeling to examine relationships among particulate matter pollution, land use, and population growth in Long Beach, California, producing 12 maps and reports.
- Created and tested a transportation engineering educational game for K–12 students, with approximately 20

participants and positive feedback outcomes.

- Contributed to the 2023 AI City Challenge Track 5 on helmet-rule violation detection using computer vision and object detection, achieving an mAP of 0.5267 and resulting in an arXiv preprint.

SELECTED PRESENTATIONS

Predicting Barge Presence and Quantity on Inland Waterways Using Vessel Tracking Data: A Machine Learning Approach

- Lectern presentation, Smart Rivers 2025, Memphis, Tennessee, Sept. 8–12, 2025.
- Poster presentation, Missouri Valley District Institute of Transportation Engineers (MOVITE) Spring Meeting, Memphis, Tennessee, 2025.
- Lectern presentation, 104th Transportation Research Board (TRB) Annual Meeting, Washington, D.C., Jan. 5–9, 2025.

Identifying Waterway Traffic Flow Patterns Using Modified Clustering

- Lectern presentation, 104th Transportation Research Board (TRB) Annual Meeting, Washington, D.C., Jan. 5–9, 2025.

Real-Time Barge Detection Using Traffic Cameras and Deep Learning on Inland Waterways

- Poster presentation, Missouri Valley District Institute of Transportation Engineers (MOVITE) Spring Meeting, Omaha, Nebraska, Apr. 17–19, 2024 - Best Graduate Student Poster Award.

Traffic Cameras to Detect Inland Waterway Barge Traffic: An Application of Machine Learning

- Lectern presentation, 103rd Transportation Research Board (TRB) Annual Meeting, Washington, D.C., Jan. 7–11, 2024.
- Poster presentation, Missouri Valley District Institute of Transportation Engineers (MOVITE) Spring Meeting, Cedar Rapids, Iowa, 2023 - Best Graduate Student Poster Award.

INVITED TALKS AND PANELS

External Invited Talks:

- Invited online seminar presentation, Lamar University, hosted by Dr. Maryam Hamidi, Feb. 13, 2025. Topic: Predicting Barge Presence and Quantity on Inland Waterways Using Vessel Tracking Data.

Departmental & Panel Presentations:

- Student panel speaker, Arkansas Chemical Engineering Graduate Students (AChEGS) Professional Development Workshop, University of Arkansas, Apr. 2, 2026. Topic: Getting an internship and succeeding during the internship.
- Invited speaker, Department of Civil Engineering, University of Arkansas, 2024. Topic: Life in graduate school as a civil engineering student at the University of Arkansas.
- Invited speaker, Department of Civil Engineering, University of Arkansas, 2025. Topic: Life in graduate school as a civil engineering student at the University of Arkansas.

FELLOWSHIPS AND PROFESSIONAL DEVELOPMENT

- Hodges Graduate Entrepreneurial Fellowship, Office of Entrepreneurship & Innovation, University of Arkansas (2026). Competitive fellowship awarded to outstanding graduate researchers demonstrating entrepreneurial potential and innovation impact; includes a \$10,000 stipend and certificate program in entrepreneurship and innovation.
- Doctoral Academy Fellow, University of Arkansas (2024–2028). Competitive merit-based fellowship awarded to outstanding doctoral researchers demonstrating exceptional academic achievement and research potential; includes a \$12,000 annual stipend over four years (\$48,000 total award).
- WODCON XXIV Fellow, Western Dredging Association (WEDA) - a selective fellowship awarded to outstanding early-career professionals for participation in the premier international dredging and waterways conference, San Diego, California, June 23–27, 2025.
- National Science Foundation (NSF) I-Corps Regional Program Participant, George Mason University, Mid-South Region Hub, Feb. 11, 2026 – Mar. 26, 2026.
- FLIPspace: Advanced Mobility Arkansas Participant, Jan. 24, 2025 – May 22, 2025.
- Recognition for Program-High 48 Customer Discovery Interviews, FLIPspace: Advanced Mobility Arkansas (2025).

HONORS AND AWARDS

Institutional and Academic Honors

- Hodges Graduate Entrepreneurial Fellowship (\$10,000), Office of Entrepreneurship & Innovation, University of Arkansas (2026) - Competitive award recognizing outstanding graduate researchers demonstrating entrepreneurial potential and innovation impact.
- Doctoral Academy Fellow (\$12,000/year; \$48,000 total), University of Arkansas (2024–2028) - Competitive merit-based fellowship awarded to exceptional doctoral researchers demonstrating outstanding academic achievement and research potential.
- Provost State and National Award Recipient (2025, 2026) - institutional recognition for academic excellence and research impact at the state and national level; received in two consecutive years.
- 2025–2026 Graduate Student Intern of the Year, College of Engineering, University of Arkansas (2026).
- Graduate Student of the Year in Service, Department of Civil Engineering, University of Arkansas (2025).
- Civil Engineering Master's Student of the Year, University of Arkansas (2024).
- Tau Beta Pi, Arkansas Alpha Chapter Engineering Honor Society (2024).
- Excellent Student Award (2021).
- Provost Excellent Student Award, KNUST (2017).
- Plan Ghana and Prudential Life Scholarship Award (2015).

Research, Paper, Poster, and Competition Awards

- 2nd Place Winner, 2026 Thomas J. Seburn Student Paper Award (2026).
- 1st Place Winner, 2025 Missouri Valley District Institute of Transportation Engineers (MOVITE) Traffic Bowl Competition (2025) - regional transportation engineering knowledge competition.
- 1st Place Winner, 2025 MOVITE Student Poster Competition (2025).
- 3rd Place Winner, 2025 MOVITE Student 5 Minute Thesis Competition (2025).
- 3rd Place Winner, 2025 Thomas J. Seburn Student Paper Award (2025).
- 2nd Place Winner, 2024 MOVITE Student Poster Competition (2024).
- 2nd Place Winner, 2024 Thomas J. Seburn Student Paper Award (2024).
- 1st Place Winner, 2023 MOVITE Student Poster Competition (2023).
- 1st Place Winner, 2023 MOVITE Traffic Bowl Competition (2023) - regional transportation engineering knowledge competition.
- Institute of Transportation Engineers (ITE) International Collegiate Traffic Bowl Grand Championship Participant (2023) - international transportation engineering knowledge competition, one of select teams to advance to the international championship level.

Service and Community Recognition

- Center for Community Engagement Service Award (2025).

PROPOSALS AND GRANTS

- Barge Detection for Situational Awareness and Commodity Monitoring to Advance Next Generation Transportation and Logistics. Submitted to the Arkansas Research Alliance, 2024. Contributed to technical writing, methodology development, and proposal framing for grant funding in support of next-generation inland waterway monitoring systems.

MENTORING

- Provided formal and informal mentoring to approximately five undergraduate and graduate students and interdisciplinary student teams in areas including career guidance, research development, and academic/professional growth.

CERTIFICATIONS AND ADDITIONAL TRAINING

- Arkansas Summer Research Institute (ASRI), Arkansas NSF EPSCoR / Arkansas School for Mathematics, Sciences, and the Arts (ASMSA).
- Engineering Project Management: Initiating and Planning, Rice University (Coursera).

TECHNICAL SKILLS

Core Technical Strengths: Python; QGIS; YOLO; PyTorch; ArcGIS Pro; scikit-learn; pandas; NumPy; OpenCV;

ONNX.

Machine Learning and Computer Vision: Object detection; YOLO; convolutional neural networks (CNNs); PyTorch; scikit-learn; OpenCV.

Geospatial and Transportation Tools: QGIS; ArcGIS Pro; ArcMap; Google Earth Pro; geospatial Python workflows; Civil 3D.

Analytics and Data Tools: SQL; PostgreSQL; Matplotlib; Seaborn; Tableau; Power BI.

Working Familiarity: Power BI; PostgreSQL; MATLAB.

Transportation Data Domains: AIS vessel tracking data; freight network data; traffic camera imagery; satellite imagery; commodity flow data.

LEADERSHIP, SERVICE, AND PROFESSIONAL ENGAGEMENT

Institute of Transportation Engineers (ITE) – University of Arkansas

- President, May 2024 – May 2026.
- Treasurer, Mar. 2023 – May 2024.

Tau Beta Pi – University of Arkansas

- Graduate Coordinator, May 2025 – Present.

Outreach and Community Engagement

- Member, iFriend Program, University of Arkansas, Jan. 2023 – Present.
- Instructional Facilitator, Department of Civil Engineering, University of Arkansas, Aug. 2023.
- Food Pantry Volunteer, The Salvation Army, Aug. 2023 – Nov. 2023.
- Cook, International Cultural Team, Aug. 2023 – Nov. 2023.
- Tree Planter, Rotaract, Sep. 2019 – Sep. 2020.

Additional Leadership and Service

- Multimedia Specialist, Trinity Fellowship, Aug. 2023 – Present.
- Blood donor, American Red Cross Society.
- Presenter, Girls Scout Thinking Day, 2023.

PROFESSIONAL MEMBERSHIPS

- Institute of Transportation Engineers (ITE), Member, Jan. 2023 – Present.
- American Society of Civil Engineers (ASCE), Member, Jan. 2023 – Present.
- Ghana Institute of Engineering (GHIE), Member, Sept. 2021 – Present.
- National Society of Black Engineers (NSBE), Member, Jan. 2018 – Present.
- Ghana Engineering Students Association (GESA), Member, Oct. 2021 – Present.